

Migration, Local Labor Demand And Urban Labor Reallocation

Special Topic 4: Urban and Labor – Week 5

Central question

- How do workers, firms, wages, rents, and commuting flows adjust when local labor demand changes?
- Migration is one margin in a spatial labor-adjustment system.
- Local shocks move people, jobs, firms, rents, commute flows, and neighborhoods.
- The key empirical task is incidence: who can adjust, who gains, and who remains exposed?

Where Week 5 fits

- Week 1: job-residence bundles define local labor opportunity.
- Week 2: housing and moving costs limit access to productive places.
- Week 3: segregation and neighborhoods create unequal job access.
- Week 4: safety and environment change the feasible set for work.
- Week 5: local shocks trigger dynamic reallocation across the whole system.

Spatial labor-adjustment value

$$V_{ijrt} = w_{jt}(D_{\ell t}) - R_{rt}(D_{\ell t}) - \tau_{jrt} + A_{rt} - M_{irt} + \varepsilon_{ijrt}$$

Term	Interpretation
$w(D)$	workplace wage response to local demand
$R(D)$	rent or housing-cost capitalization
τ	commute cost linking residence and workplace
A	amenities and disamenities
M	moving cost and attachment to place

Adjustment margins

Margin	What moves	Incidence question
Wages	pay, earnings, job quality	do workers share gains?
Employment	jobs, unemployment, LFPR	who remains attached?
Migration	inflows and outflows	who can move?
Commuting	residence-workplace flows	who accesses without moving?
Rents	housing costs, prices	who captures surplus?
Firms	entry, exit, relocation	are job ladders rebuilt?
Composition	sector, skill, neighborhood mix	who is being compared?

Migration and commuting

- Migration changes residence and the whole job-residence bundle.
- Commuting changes access to workplaces without changing residence.
- A job boom can raise workplace employment with little resident population growth.
- Commuting openness can lower wage pressure and spread gains across places.
- Housing constraints can convert labor demand into rent growth instead of worker inflow.

Incidence after a local shock

Group	Possible gain	Possible loss
Incumbent workers	higher wages, job access	rent pressure, scarring
Migrants	entry to opportunity	moving costs, crowding
Commuters	access to wage gains	time and transport costs
Landlords/homeowners	capitalization	local risk if decline
Firms	thicker labor market	higher labor and land costs
Local government	tax base	service pressure, fiscal decline

Persistence and hysteresis

- Shocks can scar workers through job loss, lost tenure, sectoral mismatch, and weaker networks.
- Local employment can recover while incumbent workers remain on lower earnings paths.
- Workers can recover by leaving even when the origin place declines.
- Place recovery and person recovery are different estimands.
- Dynamic designs must define incumbents before the shock.

Places versus people

Object	Outcomes
Place	employment, population, establishments, rents, tax base, commuting flows
Incumbent worker	earnings, employment, occupation, commute, moving cost, welfare
Migrant	destination job access, housing cost, local networks, earnings path
Commuter	workplace wage access, travel burden, residence-based welfare
Firm	hiring costs, entry, exit, productivity, location choice
Owner or landlord	capitalization, rent income, wealth exposure

Gentrification as labor restructuring

- Gentrification changes more than rents and demographics.
- It can shift local service demand, job composition, business turnover, and commute patterns.
- Incumbent residents may gain from new jobs, safer amenities, or ownership wealth.
- Incumbent renters may lose through rent pressure, network disruption, and longer commutes.
- The labor question is which jobs are created, destroyed, upgraded, or made inaccessible.

Diversification as worker insurance

- Specialized places offer upside but expose workers to sector-specific shocks.
- Diversified places may provide alternative local job ladders after downturns.
- The insurance value depends on skill transferability, housing access, and local hiring networks.
- Diversity can proxy for size, amenities, human capital, or institutions.
- Good designs distinguish resilience from selection into already-strong cities.

Methods and data

Setting	Design	Data
Local demand shocks Migration incidence	shift-share, IV, event study predicted-destination IV	ACS, QWI, industry shares IRS flows, settlement shares
Commuting access Persistence	transport or travel-time shocks dynamic local-shock panels	LEHD, LODES, GTFS matched workers, CZ pan- els
Gentrification Diversification	job-location and tract panels shock by industry mix	business data, rents, ACS sector shares, worker out- comes
Equilibrium incidence	spatial calibration	wages, rents, flows, hous- ing

What a good design shows

- The shock is separated from endogenous local trends.
- The geography matches the mechanism: residence, workplace, tract, county, CZ, or flow market.
- Wages are interpreted with rents, commuting, migration, and composition.
- Incumbents, migrants, commuters, firms, and landlords are not collapsed into one effect.
- The design states whether the target is nominal earnings, real access, resilience, or welfare.

- Reproduce: synthetic local demand shock with wages, rents, population, commuting, and firms.
- Diagnose: classify outputs as place objects or person objects and assign incidence.
- Transfer: gentrification scenario with local service demand and rent pressure.
- Transfer: diversification scenario with specialized and diversified places hit by a sector shock.
- Goal: interpret an adjustment vector before interpreting one coefficient.

Takeaways

- Migration and commuting are linked margins in local labor adjustment.
- Local shocks have incidence across workers, migrants, commuters, firms, and property owners.
- Place recovery is not the same thing as worker recovery.
- Gentrification and diversification are labor topics when they restructure jobs and risk.
- Week 6 turns these objects into research designs.